

MICHAEL BECKER
Naperville, Illinois, 60540 | 630.335.4765
Michael.S.Becker02@outlook.com | [linkedin.com/in/michael-becker2025](https://www.linkedin.com/in/michael-becker2025)

PROFESSIONAL SUMMARY

Cybersecurity Engineering student with hands-on experience in data analytics, full-stack development, and AI integration. Proven ability to build scalable systems and analyze large datasets while implementing security protocols. Seeking entry-level technology role.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, C, C++, SQL, PowerShell, Bash

Development: React.js, Spring Boot, MySQL, Docker, AWS, Git, WebSockets

Security: Splunk, Nessus, Metasploit, Wireshark, NIST Framework, ISO 27001

Machine Learning: Threat detection modeling, XGBoost, Isolation Forest, Ollama LLM

Data Analytics: Power BI, Microsoft SQL Server, statistical analysis

EDUCATION

Iowa State University, School of Engineering

Bachelor of Science in Cyber Security Engineering | December 2025

- Dean's List | Journey Scholarship Recipient
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EXPERIENCE

Consultative Analytics Intern | Broadspire | May – August 2025

- Developed Python scripts to process years of historical claim data for Microsoft SQL Server integration
- Created automated data pipelines and Power BI dashboards, improving analytical workflow efficiency
- Collaborated with engineering teams to identify patterns supporting business decision-making

Cybersecurity Research and Analysis | Iowa State University | 2023-Present

- Performed penetration testing using Splunk, Nessus, and Metasploit to identify vulnerabilities
 - Designed secure network systems with firewalls, VPNs, and IDS/IPS while conducting assessments
 - Conducted IT audits and SOC examinations for ISO 27001 and NIST framework compliance
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PROJECTS

Machine Learning Threat Detection System | Personal Project | Current

- Architected threat detection platform processing 10,000+ security events/hour using Python and Ollama LLM
- Implemented hybrid AI pipeline combining traditional ML models with large language models for automated analysis
- Built Docker containerization and AWS deployment pipeline demonstrating DevOps practices

SmartClass Web Application | Senior Design Project | Spring 2024 - Fall 2025

- Developed full-stack application using React.js frontend and Spring Boot backend with MySQL database
- Implemented real-time WebSocket communication for live classroom interaction and polling
- Designed role-based authentication system supporting multiple user types

Wireless Security Analysis | CprE 4370 | Fall 2025

- Analyzed security vulnerabilities in cellular networks, WiFi protocols, and IEEE 802.15 technologies
- Conducted penetration testing using Aircrack-ng and developed comprehensive security assessment reports

Cryptographic Implementation | CprE/CyBE 3310

- Implemented AES, RSA, and elliptic curve algorithms using Python and OpenSSL libraries
 - Designed secure key distribution and authentication systems following industry best practices
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ADDITIONAL EXPERIENCE

Additional work experience available upon request

